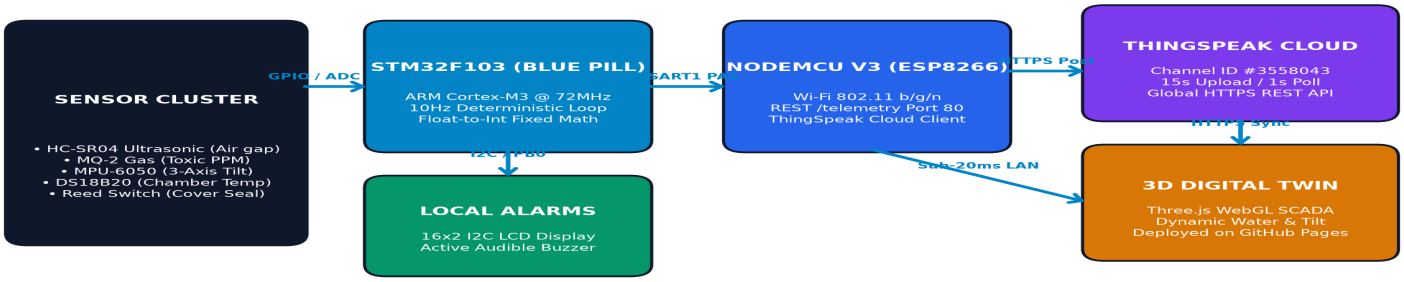
SMART MANHOLE UTILITIES URBAN IOT

Autonomous Sub-Surface Monitoring, Real-Time Cloud Telemetry & 3D Digital Twin SCADA

BATCH: BATCH 8	PROJECT GROUP: P077	2400040006 P V Kishor Kumar Reddy	2400040004 T Perumal	2400040303 Chundu Uday Sai	2400040339 Ayan Chowdhury

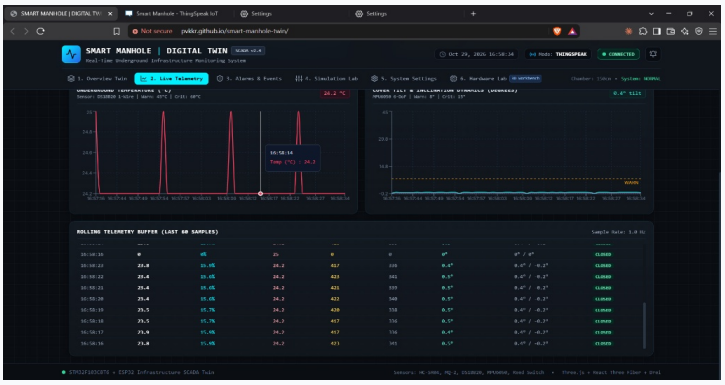
1. SYSTEM ARCHITECTURE & END-TO-END TELEMETRY PIPELINE



2. HARDWARE PROTOTYPE & 3D DIGITAL TWIN SCADA INTERFACE

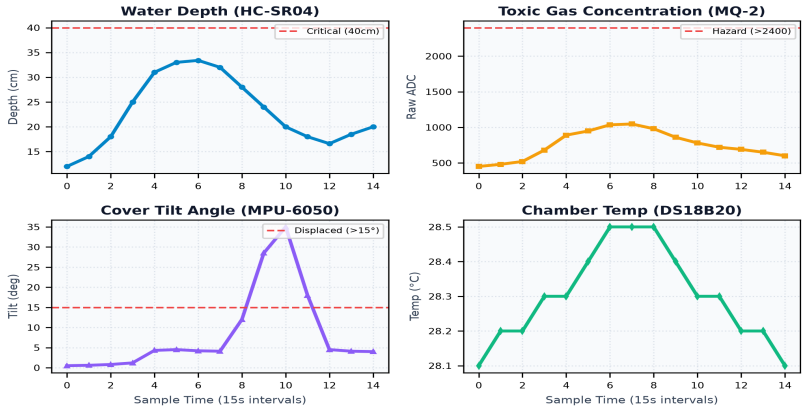


Physical Hardware Circuit: 16x2 LCD + NodeMCU V3 + STM32 Core



Live 3D Digital Twin: Physics-accurate chamber water level & lid tilt

3. MULTI-SENSOR LIVE TELEMETRY & FIELD VERIFICATION



Real-Time Multi-Sensor Telemetry Graphs (ThingSpeak Channel #3558043)

Key Project Objectives:

- **Flood Pre-emption:** HC-SR04 continuous water surcharge measurement.
- **Worker Safety:** MQ-2 toxic methane/sewer gas alert threshold (ADC >2400).
- **Anti-Tamper & Theft:** MPU-6050 3-axis tilt + Reed switch cover seal.

Engineering Results Overcome:

- **Float Fix:** Converted ARM GCC nano.specs to fixed-point (%d.%d).
- **I2C Lockup:** Eliminated bus freeze with strict I2C_Stop & restart settle.
- **Zero-Latency:** Sub-20ms local REST (/telemetry) + 1s ThingSpeak cache-bust.

Real-World Deployment:

- **Dual Core:** STM32 bare-metal deterministic loop + NodeMCU ESP8266.
- **5V Voltage Divider:** 1kΩ/2kΩ protective divider safeguarding STM32 PA2.